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Comparison of Antioxidant and Antiproliferation Activities of Polysaccharides from Eight Species of Medicinal Mushrooms

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ABSTRACT: Polysaccharides from mushrooms including *Pleurotus eryngii*, *P. ostreatus*, *P. nebrodensis*, *Lentinus edode*s, *Hypsizygus marmoreus*, *Flammulina velutipes*, *Ganoderma lucidum*, and *Hericium erinaceus* were isolated by water extraction and alcohol precipitation. Our results suggest that all tested polysaccharides have the significant antioxidant capacities of scavenging free radicals (1,1-diphenyl-2-picrylhydrazyl and hydroxyl radicals). Among them, the *H. erinaceus* polysaccharide exhibits the highest 1,1-diphenyl-2-picrylhydrazyl radical-scavenging activity, whereas the *L. edodes* polysaccharide shows the strongest scavenging ability for hydroxyl radicals. Furthermore, using the MCF-7 breast cancer cell line and HeLa cells, all 8 selected polysaccharides are able to inhibit the proliferation of tumor cells, but the strength of inhibition varied depending on the mushroom species and the concentration used. Notably, *G. lucidum* polysaccharide shows the highest inhibition activity on MCF-7 cells. By comparison, *H. erinaceus* polysaccharide has the strongest inhibitory effect on HeLa cells. Moreover, high-performance liquid chromatography with a carbohydrate analysis column showed significant differences in polysaccharide components among these mushrooms. Thus our data suggest that the different species of mushrooms have the variable functions because of their own specific polysaccharide components. The 8 mushroom polysaccharides have the potential to be used as valuable functional food additives or sources of therapeutic agents for antioxidant and cancer treatments, especially polysaccharides from *H. erinaceus*, *L. edodes*, and *G. lucidum*.

KEY WORDS: medicinal mushrooms, polysaccharides, antioxidant, antiproliferation activity

ABBREVIATIONS: DPPH, 1,1-diphenyl-2-picrylhydrazyl; **HPLC,** high-performance liquid chromatography; **MTT,** 3-(4, 5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide.

I. INTRODUCTION

As a kind of popular traditional food in Asia, mushrooms have been valued as a resource to support health for thousands of years.[1,2] Many species of mushrooms have not only biologically active compounds with antioxidant effects[3,4] but also medicinal properties in remedies for diseases.[5–7] For example, *Ganoderma lucidum* polysaccharide possesses antitumor activity,[8] whereas polysaccharide from *Pleurotus eryngii* has hepatoprotective and hypolipidemic effects,[7] and *Hericium erinaceus* polysaccharide could improve the immunity.[9] Our previous studies have verified that polysaccharides from 8 common mushrooms, including *P. eryngii*, *P. ostreatus*, *P. nebrodensis*, *Lentinus edode*s, *Hypsizygus marmoreus*, *Flammulina velutipes*, *G. lucidum*, and *H. erinaceus* have an inhibitory effect on lipid accumulation in macrophage-derived foam cells.[10] To explore the potential functions of the antioxidant and the antiproliferation activities, as well as the related composition, of above-selected 8 polysaccharides, in this study we systematically analyzed and compared the 1,1-diphenyl-2-picrylhydrazyl (DPPH) and hydroxyl radical scavenging activities as well as antiproliferation abilities against breast cancer cell line MCF-7 and HeLa cells *in vitro*. Consequently, we found that the 8 selected polysaccharides might be good potential

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candidates as antioxidant and anticancer food additives, especially polysaccharides from *H. erinaceus*, *L. edodes*, and *G. lucidum*.

II. MATERIALS AND METHODS

A. Materials and Reagents

The mushroom fruiting bodies of *P. eryngii*, *P. ostreatus*, *P. nebrodensis*, *L. edode*s, *H. marmoreus*, *F. velutipes*, *G. lucidum*, and *H. erinaceus* were purchased from Dashan Group Ltd. Co. of China. The chemical reagents used in this study were of analytical grade.

B. Extraction and Preparation of the Crude Polysaccharides

The crude polysaccharides from 8 mushroom fruiting bodies were extracted and prepared as previously described.[11] Briefly, the dried fruiting bodies were pulverized in an electric mill to powder of a size that was able to pass through a 0.425-mm sieve, then the powder was extracted with hot water (90°C) for 3 hours and filtered with filter paper. The filtrate was concentrated under reduced pressure in an evaporator and precipitated by adding ethanol at a final concentration of 80% at 4°C overnight. The precipitate was dissolved in distilled water at a concentration of 50 mg/mL; then 10 mg/mL of trypsin (pH 8.1) was added to a ratio of 1:0.4 (v/v) at 37°C for 3 hours to remove the proteins and mixed with 30% hydrogen peroxide at 1:0.2 (v/v) to destain the pigments. Last, 80% ethanol (final concentration) was added to the solution and kept at 4°C overnight. The precipitate then was collected and dried to form a water-soluble crude polysaccharide extract.

C. Detection of Radical-Scavenging Activities

Polysaccharide ability to scavenge DPPH radicals was measured according to the method described by Yamaguchi et al.,[12] with some modifications. Various concentrations of crude polysaccharides were mixed with 0.5 mL of ethanol solution containing 0.1 mmol/L DPPH radicals. The mixture was shaken vigorously and left to stand for 20 min at room temperature in the dark. The reduction of DPPH radicals was determined by measuring the absorption at 517 nm. The hydroxyl radical-scavenging activity of crude polysaccharides were measured according to a previously described method,[13] with some modifications. Various concentrations of crude polysaccharides first were mixed with 1 mL of 6 mM ferrous sulphate and 6 mM salicylic acid, and then the solution was mixed with 1 mL of 6 mmol hydrogen peroxide and incubated for 1 hour at 37°C. The reduction of hydroxyl radicals was determined by measuring the absorption at 510 nm. All tests and analyses were carried out in triplicate. Radical scavenging activity was calculated with the following equation:

$$\text{Scavenging activity (\%)} = [1 - (A_x - A_{x0})/A_0] \times 100,$$

where A_x is the absorbance of sample, A_{x0} is the absorbance of the blank, and A_0 is the absorbance of the control.

D. Cell Culture and Cell Viability Assay

The human MCF-7 breast cancer cells and HeLa cells were cultured in Dulbecco's modified Eagle medium (glutamine, high glucose) supplemented with 10% fetal bovine serum (Wisent Ltd. Co., Canada) at 37°C, 5% carbon dioxide. Cell viability was measured using MTT (3-[4,5-dimethyl-thiazol-2-yl]-2,5-diphenyltetrazolium bromide) assays according to a previously described method.[14] Briefly, cells were harvested and plated in a volume of 100 µL (1000 cells/well) in 96-well microtiter plates using a multichannel pipette. The plates were incubated at 37°C for 1 hour to allow cells to reattach and equilibrate. After removing the medium, 180 µL fresh medium with crude polysaccharides from mushroom fruiting bodies (concentrations of 50, 100, and 200 µg/mL) were added, and then the cells were incubated. After 24 hours, 20 µL MTT (Sigma) solution (5 mg/mL in phosphate-buffered saline) was added and the cells continued to be in-

cubated for 4 hours. After removing the medium, 200 µL dimethyl sulfoxide were added to each cell volume and the plates were shaken for 10 min to solve the crystal. The plates were read at 570 nm using a microplate reader (SpectraMax M2; Molecular Devices, USA). All treatments were carried out in triplicate.

E. Polysaccharide Components Analysis by High-Performance Liquid Chromatography

The crude polysaccharides from mushroom fruiting bodies (10 mg) were dissolved in distilled water (2 mL) and filtered by a 0.22-µm filter. Then the flow-through solution was analyzed by a high-performance liquid chromatography (HPLC) instrument (Agilent 1100) with a carbohydrate analysis column (300 × 7.8 mm; Bio-Rad Aminex HPX-87H), using a mobile phase condition (5 mmol/L sulfuric acid at a rate of 0.5 mL/min) and a refractive index detector.

F. Statistical Analysis

Results were expressed as means ± standard deviations. SPSS 19.0 was used for all analyses.

III. RESULTS

A. Antioxidant Activity of Polysaccharides *In Vitro*

The crude polysaccharides from selected mushroom fruiting bodies were obtained by water extraction and alcohol precipitation. To get rid of some small molecules interfering in the function of polysaccharides, crude polysaccharides were dialyzed using a 3.5-kD dialysis membrane. Consequently, no obvious absorption peak was observed at 260 or 280 nm (Fig. 1) through wavelength scanning from 190 to 490 nm, suggesting no detectable protein or nucleic acid existed in crude polysaccharides.

Hydroxyl radicals are the most harmful reactive oxygen species, and they are mainly responsible for oxidative injury to biomoleculed.[15] To detect whether these polysaccharides have the antioxidant function of scavenging free radicals, we investigated the scavenging ability of these crude polysaccharides on DPPH radicals and hydroxyl radicals *in vitro*. As shown in Fig. 2, all polysaccharides exhibited effective DPPH radical-scavenging activity, which was enhanced with corresponding increases in concentration. Interestingly, when polysaccharide concentrations of 50 or 100 µg/mL were used, the polysaccharide from *L. edodes* had the strongest DPPH-scavenging activity, which was 52.6% and 68.5%, respectively. When the polysaccharide was at a concentration of 200 µg/mL, however, the DPPH-scavenging activity of *H. erinaceus* polysaccharide reached 76.9%, which was the highest among the 8 tested polysaccharides. Our results demonstrate that polysaccharides of these 8 selected mushrooms all have DPPH-scavenging activity, but polysaccharides from various types of mushrooms have different scavenging activity for DPPH, and the scavenging activity of *H. erinaceus* polysaccharide was the highest.

Next, the scavenging effects of 8 crude polysaccharides on hydroxyl radicals were investigated. As a result, all 8 crude polysaccharides exhibited the hydroxyl radical-scavenging activity in a dose-dependent manner (Fig. 3). Based on the comparison of the scavenging rate for hydroxyl radicals at a polysaccharide concentration of 200 µg/mL, the polysaccharide of *L. edodes* had the highest hydroxyl radical-scavenging activity (59.9%). The hydroxyl radical-scavenging activity, arranged from high to low, was in the following order: *L. edodes*, *H. erinaceus*, *P. ostreatus*, *P. eryngii*, *P. nebrodensis*, *H. marmoreus*, *F. velutiper*, *G. lucidum* (Fig. 3). It indicates that all 8 tested polysaccharides have hydroxyl radical-scavenging ability, but the scavenging strength varied among them and the polysaccharide of *L. edodes* possessed the highest hydroxyl radical-scavenging capacity.

B. Antiproliferation Activity of Polysaccharides on Tumor Cell Lines

To study whether these polysaccharides have growth-inhibitory activity *in vitro*, human cancer

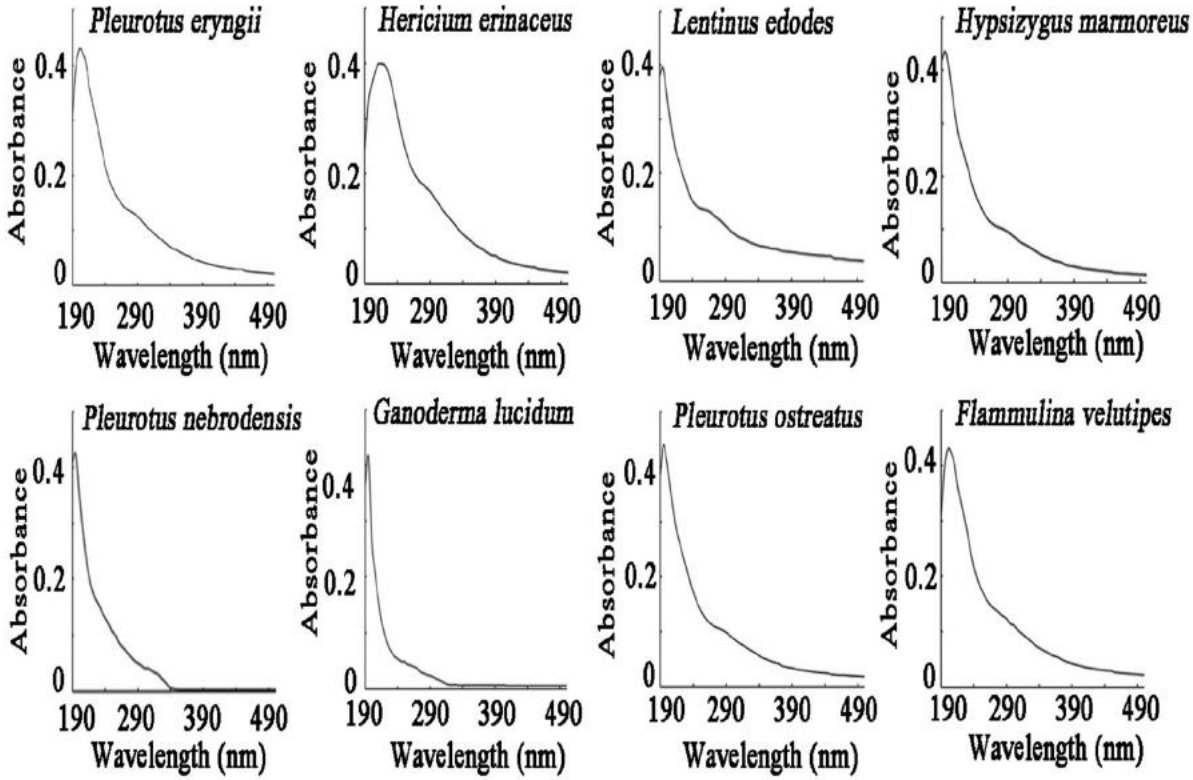

FIG. 1: Wavelength scanning diagrams of polysaccharide extracts from 8 species of medicinal mushrooms analyzed using a spectrophotometer scanning from 190 to 490 nm and showing no obvious absorption peak at 260 or 280 nm.

cell lines (MCF-7 breast cancer cells and HeLa cells) were incubated in medium with a polysaccharide concentration of 50, 100, and 200 µg/mL. Using the MTT assay, the potentially toxic effect of the polysaccharides on cancer cells was assessed. Consequently, all tested polysaccharides showed inhibitory activity on both of these cancer cell lines at polysaccharide concentrations ranging from 50 to 200 µg/mL (Figs. 4 and 5). However, inhibition rates were different. Interestingly, the polysaccharide from *G. lucidum* exhibited the highest inhibition rate on the MCF-7 cancer cells (Fig. 4), which reached 65.96% at a polysaccharidic concentration of 200 µg/mL. This suggests that the polysaccharide from *G. lucidum* has the strongest antiproliferation activity on the MCF-7 cancer cells among the 8 tested samples. In addition, the polysaccharide from *H. erinaceus* showed the highest inhibition rate (41.25%) against HeLa cells, whereas the polysaccharide from *G. lucidum* (36.59%) was next most effective against HeLa cells, both at a polysaccharide concentration of 200 µg/mL (Fig. 5). Thus, among these tested polysaccharides, the polysaccharide from *G. lucidum* probably could be used as the most effective antiproliferation candidate for the MCF-7 cancer cells and polysaccharide from *H. erinaceus* for HeLa cells.

C. Component Profile of Polysaccharides Isolated from Mushrooms

To analyze further the polysaccharide components and the polysaccharide content in these mushrooms, an HPLC system with a carbohydrate analysis column was used. Because the different retention time of HPLC represents the specific component

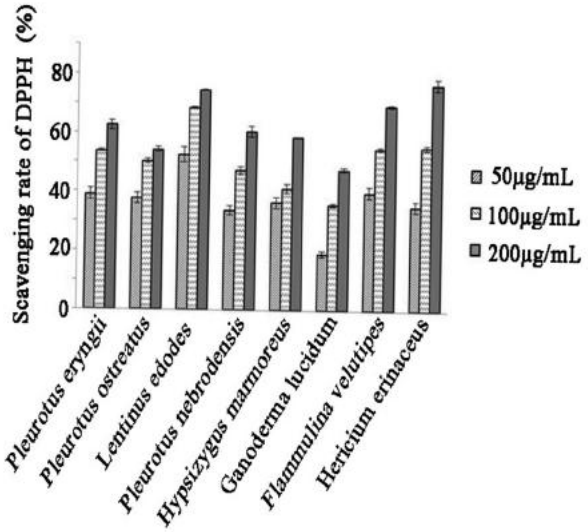

FIG. 2: DPPH radical-scavenging rates of polysaccharides from selected species of medicinal mushrooms. The polysaccharides were at different concentrations: 50, 100, and 200 µg/mL. Data represent the means ± standard deviations from 3 independent experiments.

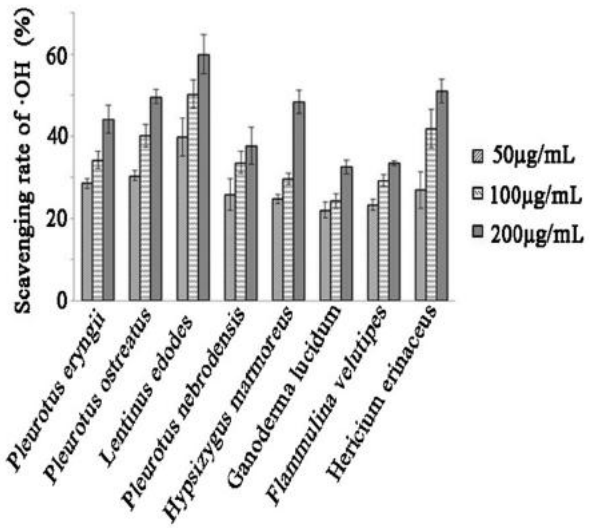

FIG. 3: Hydroxyl radical-scavenging rates of polysaccharides from 8 species of medicinal mushrooms. The polysaccharides were at different concentrations: 50, 100, and 200 µg/mL. Data represent the means ± standard deviations from 3 independent experiments.

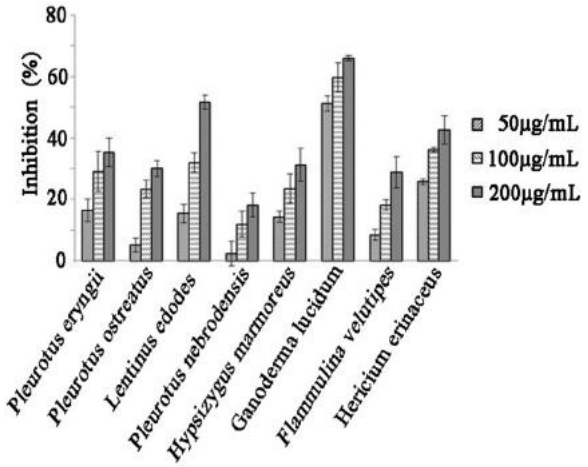

FIG. 4: Comparison of the effect of inhibition rates on the proliferation of MCF-7 cells. The MCF-7 cells were treated with different concentrations of polysaccharides: 50, 100, and 200 µg/mL. The data are presented as means ± standard deviations from 3 independent experiments.

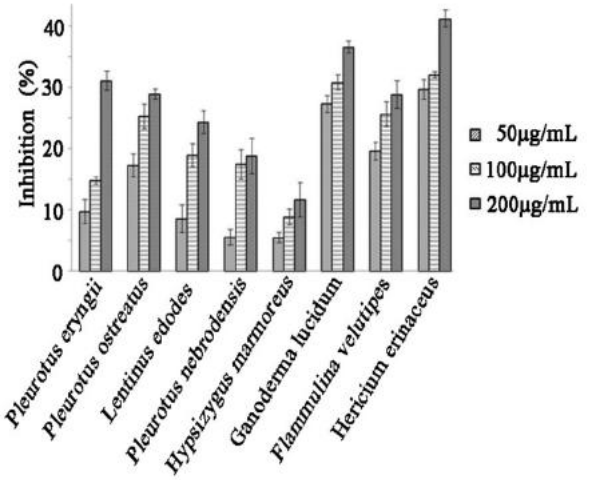

FIG. 5: Comparison of the effect of inhibition rates on the proliferation of HeLa cells. The HeLa cells were incubated with different concentrations of polysaccharides: 50, 100, and 200 µg/mL. The data are presented as means ± standard deviations from 3 independent experiments.

and the peak area is able to indicate the content of polysaccharides, the characteristics of polysaccharide fractions could be evaluated according to the peak characteristics of HPLC. As shown in Table 1, 6 major fractions contained in each crude polysaccharide were found. Interestingly, the polysaccharide fraction appeared after about 8.9 min of retention time was included in all of the tested mushrooms, but its content was significantly different, ranging from 6.39% to 55.94%. In addition,

TABLE 1: Polysaccharide Components Derived from Studied Medicinal Mushrooms, as Detected by High-Performance Liquid Chromatography

Polysaccharide Component	Species															
	Pleurotus eryngii		*Lentinus edodes*		*Pleurotus ostreatus*		*Pleurotus nebrodensis*		*Hypsizygus marmoreus*		*Flammulina velutipes*		*Ganoderma lucidum*		*Hericium erinaceus*	
	Time (min)	Peak Area (%)	Time (min)	Peak Area (%)	Time (min)	Peak Area (%)	Time (min)	Peak Area (%)	Time (min)	Peak Area (%)	Time (min)	Peak Area (%)	Time (min)	Peak Area (%)	Time (min)	Peak Area (%)
1	4.520	4.03	7.184	6.82	7.169	3.39	7.143	32.59	7.170	6.38	12.945	44.42	4.235	14.43	4.035	3.25
2	7.444	60.23	7.430	7.70	7.649	4.44	8.922	51.27	7.669	3.71	8.927	17.07	6.822	10.25	7.420	5.83
3	8.170	4.68	8.922	29.19	8.915	55.94	9.472	2.38	8.926	50.73	7.148	7.61	8.935	7.48	8.931	6.39
4	8.926	8.73	9.481	3.13	9.484	4.92	9.819	1.32	9.508	5.78	11.817	7.46	10.645	6.71	11.835	12.59
5	9.471	9.01	11.885	29.52	10.630	4.18	10.636	3.98	10.642	4.15	7.432	4.60	11.849	8.38	12.959	49.75
6	9.819	3.89	12.937	12.94	11.856	12.21	11.827	6.98	11.884	17.21	9.501	3.45	12.950	6.60	13.837	6.23
Total		90.57		89.3		85.08		98.52		87.96		84.61		53.85		84.04

the polysaccharide fraction appearing after about 12.9 min of retention time was included in only *L. edodes*, *F. velutipes*, *G. lucidum*, and *H. erinaceus*, with content varying from 6.6% to 49.75%. Moreover, *P. eryngii*, *L. edodes*, *F. velutipe*s, and *H. erinaceus* contained a polysaccharide fraction that appeared after about 7.4 min of retention time, but the content of this polysaccharide fraction ranged from 4.6% and 60.23%. Most interestingly, *G. lucidum* included a main specific polysaccharide fraction (at 4.235 min of retention time) that was completely absent in all other tested mushrooms. These data imply that each mushroom species has its own specific polysaccharide components.

IV. DISCUSSION

Free radicals and active oxygen play important roles during the process of aging and the pathogenesis of many human diseases such as cancer and atherosclerosis.[16–19] Therefore, focus on the functional foods with the effect on scavenging free radicals and antiproliferation activity for the cancer cells are increasing. Multiple lines of previous evidence have demonstrated that crude polysaccharides from edible mushrooms have the potential to prevent or attenuate human diseases including cancers, inflammatory diseases, hyperlipidemia, diabetes, and atherosclerosis.[2,20] Thus polysaccharides and polysaccharide–protein complexes isolated from mushrooms have been used as a source of therapeutic agents because of their immunomodulatory and antitumor effects.[21,22] However, which species of mushrooms have the best effect on scavenging free radicals and antiproliferation is still unknown. Our study focus on the evaluation of antioxidant and antiproliferation activities of the crude polysaccharides from 8 selected mushrooms, including *P. eryngii*, *P. ostreatus*, *P. nebrodensis*, *L. edodes*, *H. marmoreus*, *H. erinaceus*, *G. lucidum*, and *F. velutipes*. In our results the crude polysaccharides of all the 8 mushroom species exhibited a remarkable antioxidative effect against DPPH and hydroxyl radicals and an inhibitory effect on MCF-7 and HeLa cells, but this effect varied depending on the mushroom species and the polysaccharide concentration used. Moreover, HPLC analysis suggests that the different species of mushrooms have their own specific polysaccharide components, which resulted in some changes to their antioxidant and antiproliferation activities.

Our data showed that all 8 polysaccharides tested possess antioxidant ability against DPPH and hydroxyl radicals in a dose-dependent way, which is consistent with previous reports.[23,24] However, the radical-scavenging rates of the 8 polysaccharides tested have remarkable differences (Figs. 2 and 3). Thus the specific candidate could be chosen as the most effective one for scavenging various free radicals. For example, for scavenging DPPH radicals, we could choose polysaccharide from *H. erinaceus* as the most effective therapeutic agent, whereas the polysaccharide from *L. edodes* can be selected as the most useful reagent for scavenging hydroxyl radicals. This raises the question of the structural characteristics of polysaccharides from *H. erinaceus* or *L. edodes* behind the antioxidant effects. *H. erinaceus* polysaccharide was reported to be composed mainly of glucose (68.4%), arabinose (11.3%), and xylose (7.8%)[23] and has a main chain composed of β-(1→3)-linked D-glucopyranosyl residues with single-unit glucosyl branches attached to O–6 of every third backbone residue.[25] For the *L. edodes* polysaccharide, antioxidant properties were determined and the major monosaccharide compositions were D-galactose, D-glucose, and D-mannose,[24] and the *L. edodes* polysaccharide consists of glucose with a structure of β-(1→3)-D-glucose main chains and β-(1→6)-D-glucose side chains.[26] In addition, our HPLC data showed that structural characteristics of polysaccharides differ markedly among them. Specific polysaccharide components possibly lead to variable antioxidant capacities in these species of mushrooms. Taken together, our findings revealed all 8 polysaccharides tested could be used as good pharmaceutical candidates for scavenging free radicals, especially those from *H. erinaceus* and *L. edodes*.

In addition, polysaccharides from many species of mushrooms, such as *P. eryngii*, *L. edodes*, *G. lucidum*, and *H. erinaceus*, have antitumor

activity in cancer cell lines or animal experiments.[8,27–29] In our study, however, all 8 polysaccharides exhibited concentration-dependent inhibition abilities *in vitro* against human breast cancer MCF-7 cells and human cervical carcinoma HeLa cells, and inhibition rates are significantly different among the 8 polysaccharides (Figs. 4 and 5). The polysaccharide from *G. lucidum* is the best candidate against MCF-7 cells, whereas the polysaccharide of *H. erinaceus* is the most effective in suppressing proliferation of HeLa cells. Thus the polysaccharide should suit the remedy for the case, especially for different cancer treatments. Previous study has indicated that both 1,3-β-D-glucosidic and 1,4-α-D-galactosiduronic linkages existed in *G. lucidum*.[30] Notably, in our HPLC assay, *G. lucidum* contains its own special component (at about 4.235 min of retention time) as a major polysaccharide component that is not included in the other 7 mushrooms tested. This special component (at about 4.235 min of retention time) in *G. lucidum* may lead to the best effect against MCF-7 cells. As shown in Table 1, among the selected mushrooms, *H. erinaceus* includes one polysaccharide component (at about 12.9 min of retention time) in the largest amount (49.75%). This polysaccharide component could have a biologically active antioxidant capacity and antiproliferation activity so that the polysaccharide from *H. erinaceus* has the highest effective ability for scavenging DPPH radicals and inhibiting the growth of HeLa cells at a concentration of 200 μg/mL. Therefore, future work is necessary to characterize the bioactive components in these mushrooms and the detailed mechanisms behind them.

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